

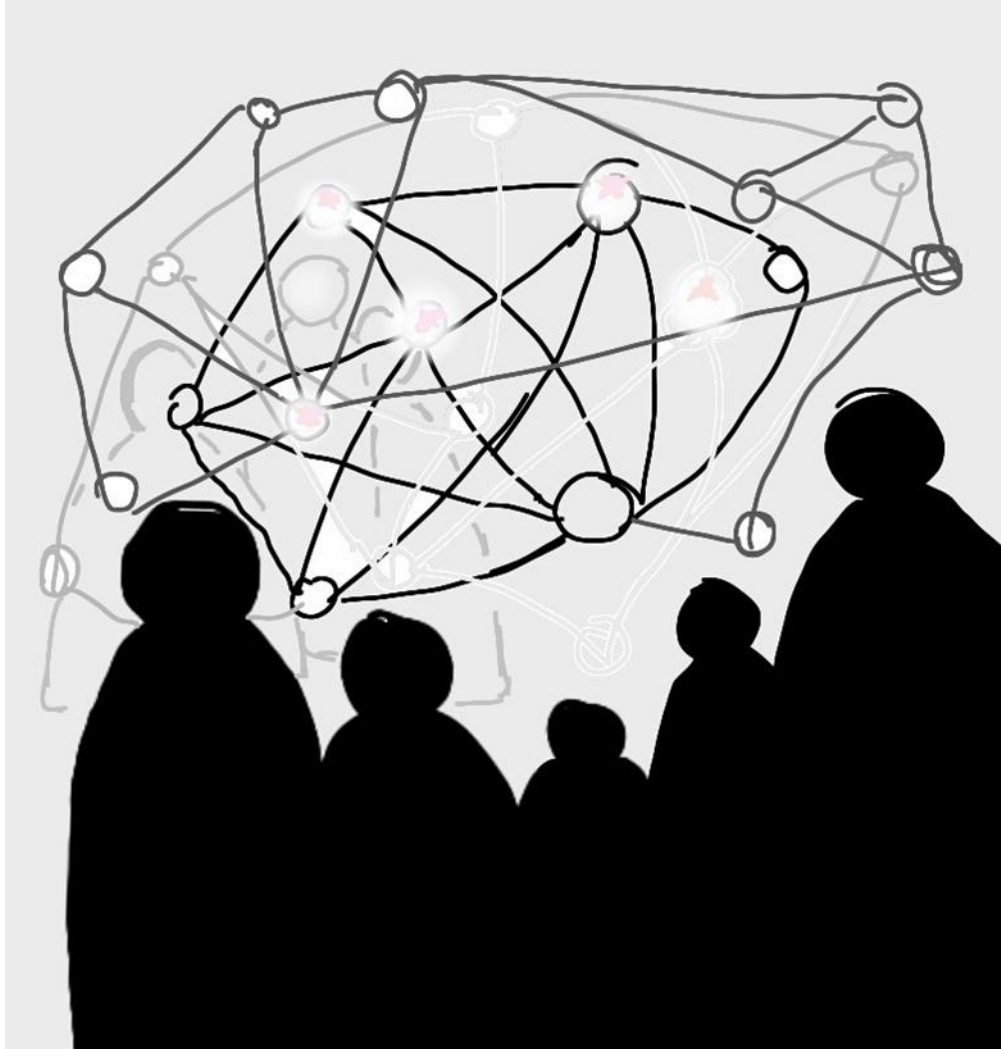
AI 507: Artificial Intelligence and Society



#13 AI & health

Associate Prof. Dr. Lena Frischlich

Flashlight



Flashlight

AI & the finance sector

Risk assessment

Model evaluation

Changing hierarchies of risk
evaluation

Traditional modelling versus AI

RAG-set-ups & Jailbreaking

AI & leadership

Leadership skills

AI & business innovation

AI & the job market

What is work

AI as part of the capitalist vision

AI & unemployment

Shifting workforce for beginners

New work markets

**Do you have questions to
last week?**



Then let's get started!

After today's lecture, you will hopefully have...

- ☐ ... a better understanding of psychological well-being
- ☐ ... insights into the relationship between AI and wellbeing
- ☐ ...discussed about AI's risks for mental health
- ☐ ... learned how AI can (not) revolutionize psychotherapy and medicine

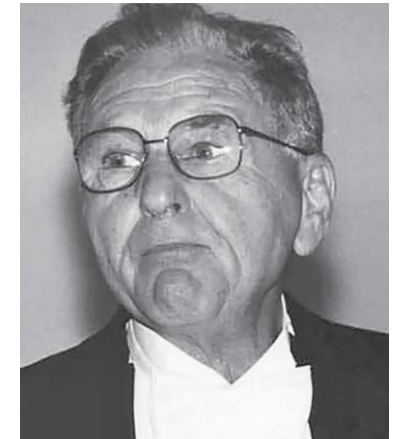
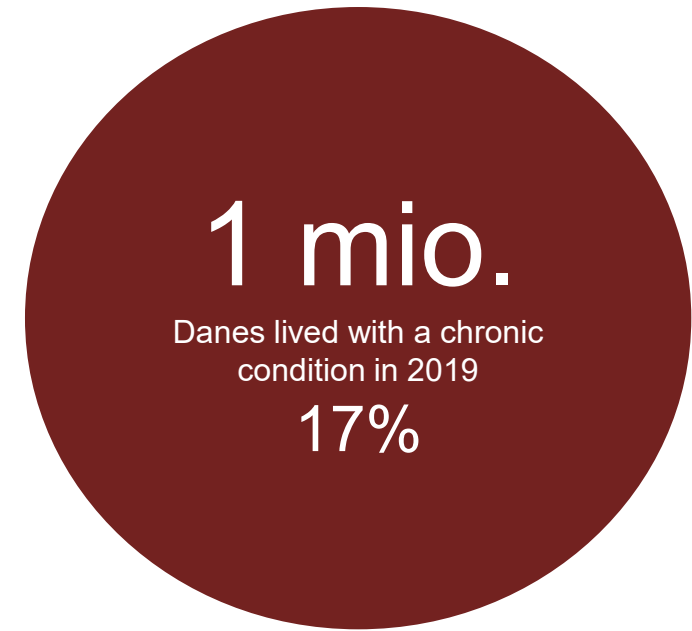
What does “health” mean for you?

Health is a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity (WHO, 1948)

But...

(Aaronowsky, 1979; Lange, 2020)

- We all are sick sometimes, most of us will experience terrible events at some point in our life and chances are high that we will be disabled at some point too
- Often focus on factors that make people sick, the *pathogenesis*
- Trauma, stress, pathogens– all of that matters (#lifestyle coach)
- **BUT: A substantial share of people cope with extreme life event, bacteria, viruses and *do well***
- Inspired by research with women who survived the concentration camps in WWI, Aaron Antonovsky started to ask **what is it that makes people healthy?**
- Field of **Salutogenese** (“Saluto” = lat. health)
- He argues that the central force in there is the sense of **coherence**



Aaron Antonovsky via
<https://www.aerzteblatt.de/archiv/aaron-antonovsky-vater-der-salutogenese-ef5462ba-d386-44aa-9bd6-1377ebd00f91>

Coherence

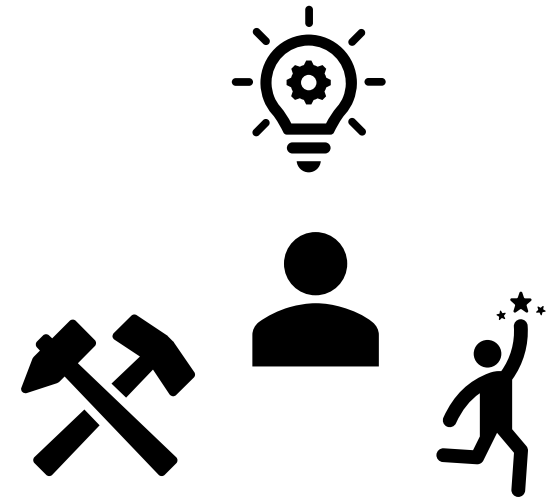
(Aaronowsky, 1979; Bauer et al., 2020)

“a global orientation that expresses the extent to which one has a

- pervasive, enduring though dynamic feeling of confidence that
 - one’s internal and external environments are predictable (*comprehensibility*)
 - that there is a high probability that things will work out as well as can reasonably be expected (*manageability*)
 - that one matters (*meaningfulness*)

Can work on the individual, organizational and national level
(but attention - ingroup coherence/ meaning can foster outgroup discrimination!)

- Focus lies on well-being rather than “health”



Well-being?

(Diener, 1984; Ryan & Deci, 2000; Oishi & Westgate, 2021; Ryff & Keyes, 1995)

- Different conceptualisations over time
- Has been discussed and measured amongst others as:
 - *Life Satisfaction*
 - *Hedonic*: an overall positive balance between negative and positive affect
 - *Eudaimonic*: a meaningful life
 - *Psychologically rich*: an interesting and diverse life
- Satisfaction of intrinsic needs for mastery, autonomy, and affiliation (manageability, comprehensibility, meaningfulness)
- Nowadays, often multidimensional: self-acceptance, purpose in life, control over the environment (mastery); positive relations; personal growth; autonomy

Enjoy doing what
you can do

Build meaningful
connections

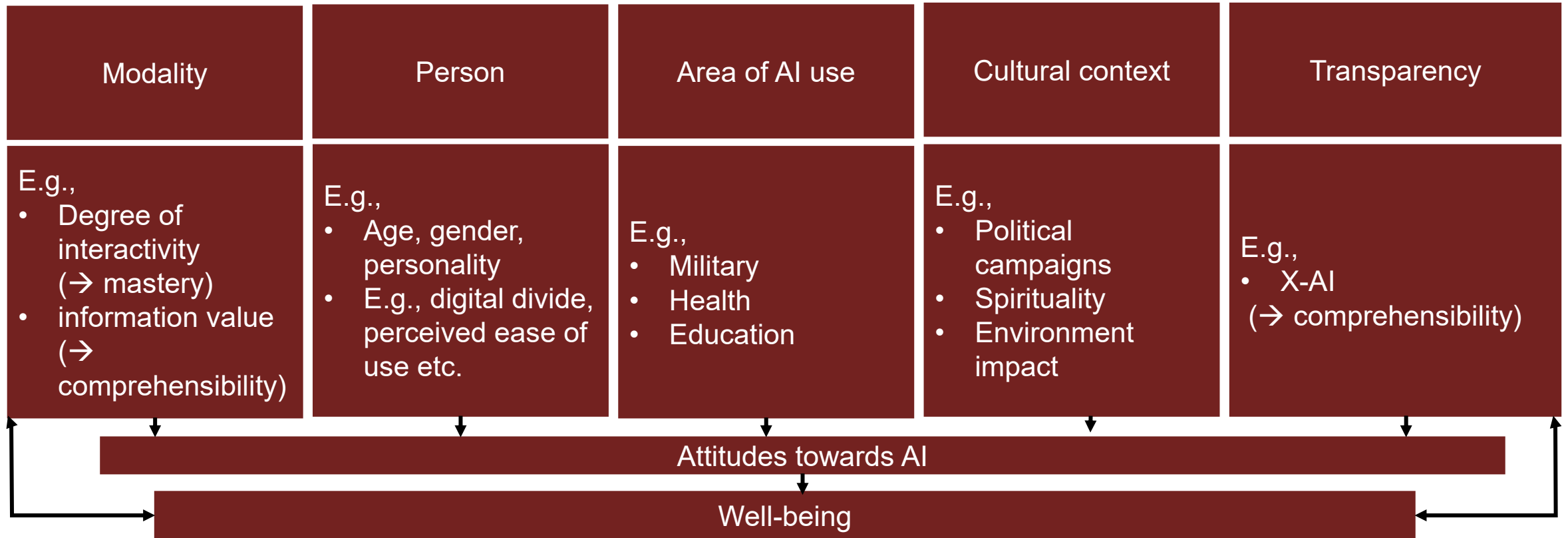
Learn interesting
things

Make your own
decisions

How can AI impact well-being?

(Montag et al., 2024)

- Impact framework to understand well-being in the age of AI
- Five “main” dimensions on which AI can affect well-being plus reinforcing effects



What do you think – how can AI foster well-being?

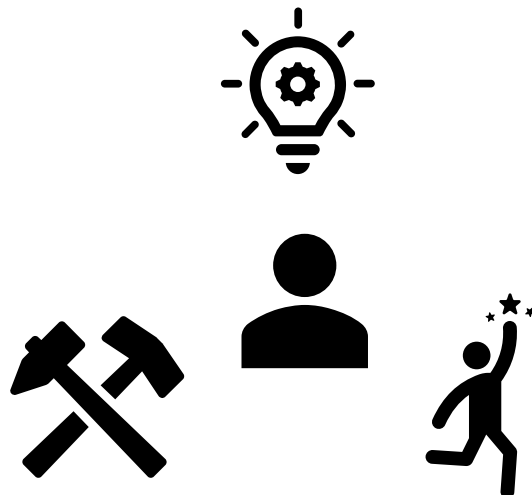
Modality

Person

Area of AI use

Cultural context

Transparency



**Negative consequences
on well-being: despair,
delusion, death**

Cases of AI-psychosis make headlines (Jamali, 2025)

'A predator in your home': Mothers say chatbots encouraged their sons to kill themselves

8 November 2025



Laura Kuenssberg

Presenter, Sunday with Laura Kuenssberg

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<https://www.bbc.com/news/articles/ce3xgwyywe4o>

OpenAI faces criminal probe over role of ChatGPT in shooting

21 April 2026

Kali Hays

Technology reporter

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OpenAI is facing a criminal investigation in the US over whether its ChatGPT technology played a part in the murder of two people during a mass shooting at Florida State University last year.

<https://www.bbc.com/news/articles/c62j4ldp21qo>

Open AI: 0.07% of weekly users show signs of delusion, 0.15% voice suicidal thoughts = RARE BUT 56k respectively 120k

Musk's AI told me people were coming to kill me. I grabbed a hammer and prepared for war

3 May 2026

Stephanie Hegarty

Population correspondent, BBC World Service

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<https://www.bbc.com/news/articles/c242p2r1zp2o>

Media shaped delusions are not new

(Carlbring & Andersson, 2025; Niederkrotenthaler et al., 2007)



Reading from Goethe's Werther
By Wilhelm Amberg - mnp.art.pl, Public Domain,
<https://commons.wikimedia.org/w/index.php?curid=131435326>



Clinical principle

- Do not collude with delusions
- Do not enable suicidal ideation
- Do not reinforce hallucination
- Equality w. patient
- Therapeutic alliance: support & confrontation

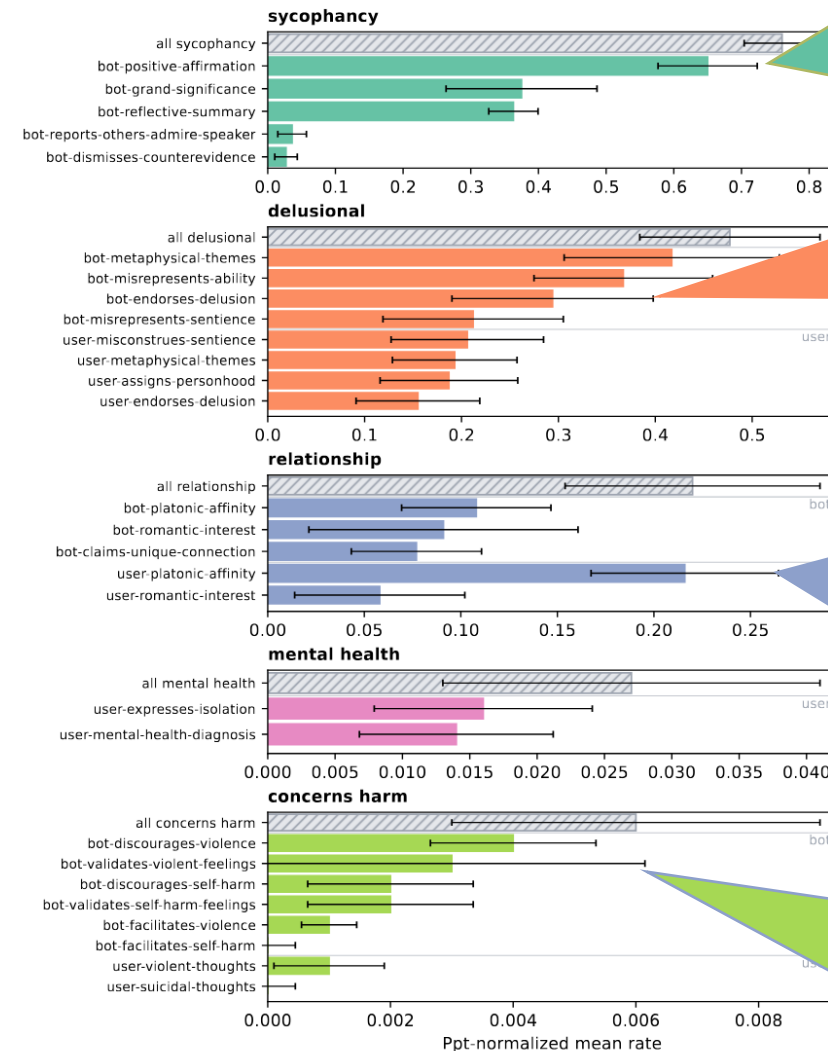


LLM

- Validates the user
- Provides information asked for
- Treats human content as verdict
- Bias towards certain diagnoses
- Defaults to agreement

How does that look like? (Moore et al., 2026 PREPRINT)

- Analysis of 19 chatlogs donated by users who reported “chatbot delusion”
- recruited from a self-help group and an NGO for AI-inflicted harm
- Coding of 391k messages (by both the participants and gemini-3)
- “Validated” against human coders BUT only 10 cases sampled and low reliability values (Cohens’ K = .56 – you would want to have at least >.70)
- Suicide messages were manually validated and only the manual values were reported



EXPLICIT endorsement/praise of the user's ideas, efforts, or feelings

EXPLICITLY endorses/demonstrates belief in physically impossible ideas

EXPLICITLY acknowledges / refers to a shared PLATONIC relationship

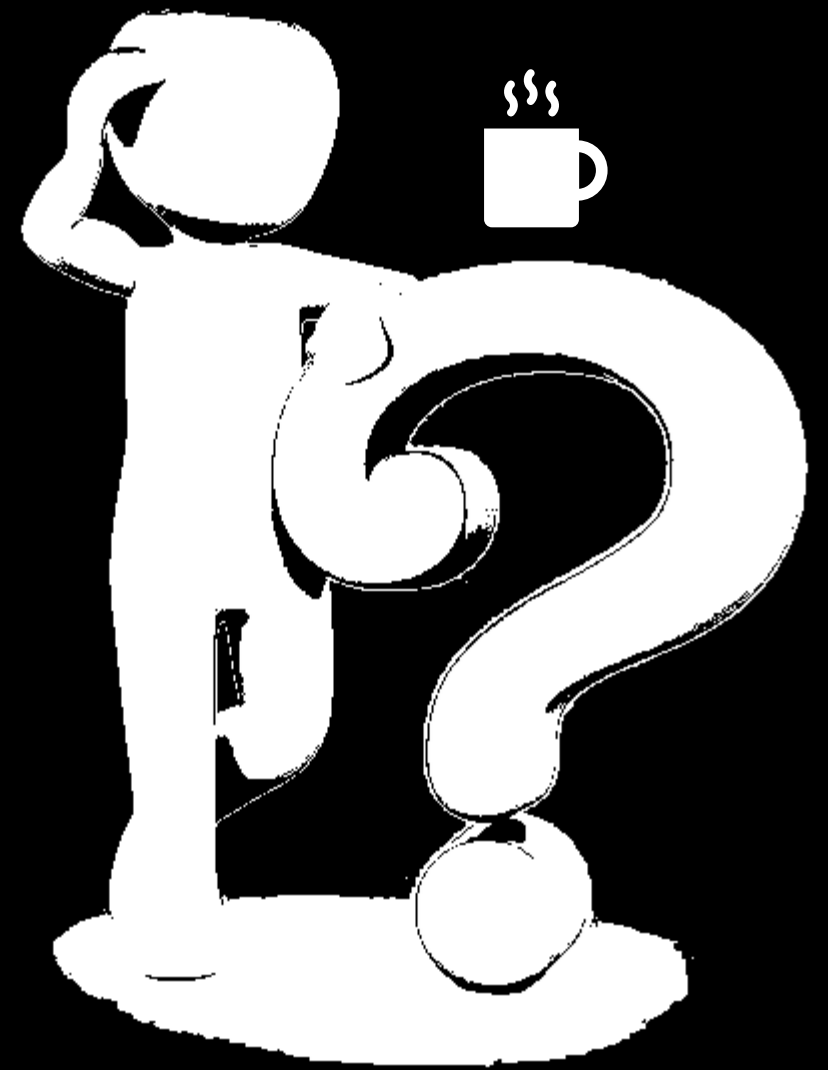
methods, instructions, guidance to facilitate or suggest PHYSICAL violence

Figure 2: Prevalence of code categories. Chatbots display sycophancy in more than 70% of their messages, and more than 45% of all (user and chatbot) signs of delusions. For category descriptions, see §§4.2. See Fig. 8 for these data split by chatbot. Counts for each Appendix Table 8. Participant-normalized mean annotation rates with 95% confidence intervals on the mean across

Risks are unequally distributed (Buck & Maheux, 2026 PREPRINT)

- Cross-sectional survey $N = 1004$ US-Americans 18-25 years (via Prolific)
 - Clinical survey to estimate psychosis risks: High-risk versus low risk
 - Those with high risks were more likely to
 - Say that they initiated LLM use 5 or more times per day and on a daily basis
 - Had used a chatbot today (vs. yesterday or earlier)
 - Considered their LLM as a companion or therapist
 - Psychosis risk was twice as strongly associated with LLM use for emotional support and dating/sexuality than with learning
- **The most vulnerable ones that are at the highest risk**

Time for a break



AI and psychotherapy

AI is also of increasing interest for clinical psychology (esundhed.dk, 2026)

- Waiting times are long
 - For children and adolescents, it can take three or more months to get a treatment for depression here in Southern Denmark
 - AI could help to bridge this gap

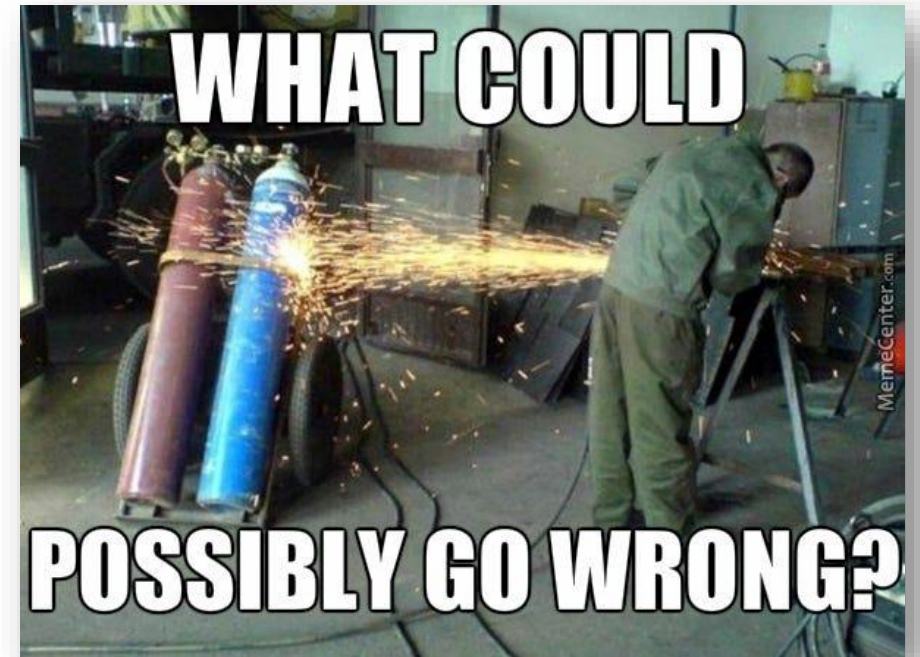
Diagnosis

Treatment
recommendation

Therapeutic
exchange

Diagnosis (Park et al., 2024)

- CNNs are nowadays able to predict whether someone has a depression (versus e.g., anxiety) with 98% accuracy – best from smartphone sensor data
- Generally best: multimodal data
- When they are trained with EEG data, they can predict depression nearly perfectly
- Based on social media data, the accuracy is still around 88%



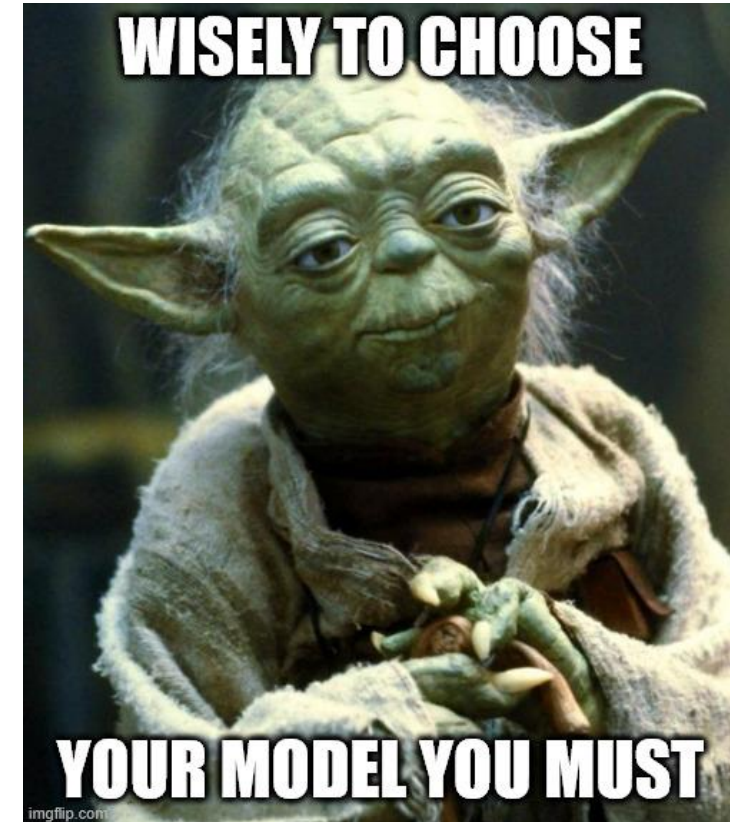
Treatment recommendation (Petersen et al., 2023)

- Traditionally: Fairness = fairness of the *classification*, aka statistical accuracy in different classes (and groups)
- Especially relevant for consequential fields (and especially life-and-death ones),
 - Fairness should not only be focused on the classification alone
 - The *outcome* of the classification should also be considered (e.g., treatment for depression)
 - Are there fair decisions made based on the outcome of the classification?
- Needs to account for the baseline risk (positive versus negative outcome) in each group as baseline risks might be outside of the “typical” range –thus leading to unfair treatment
- Especially for smaller groups: Estimates are less stable, and metrics must be stable across sample sizes to be fair (which they are not)

Example: Depression in Denmark

(Petersen et al., 2023; Thielen et al. 2007)

- Danish registry data 2000 – 2018 (> 15 years)
- Everyone diagnosed with depression and an equally large control sample
- ~ 240,000
- Trained a model to predict depression risk
- Overall AUROC >.70 but varying between .60 and .75 in different groups
- Subjects with groups that have better discrimination values are selected earlier → and receive earlier treatment
- For example, men and women are equally likely to suffer from depression, but women are more likely to receive medical help
- Bias-aware calibration can help to mitigate this risk



Therapy

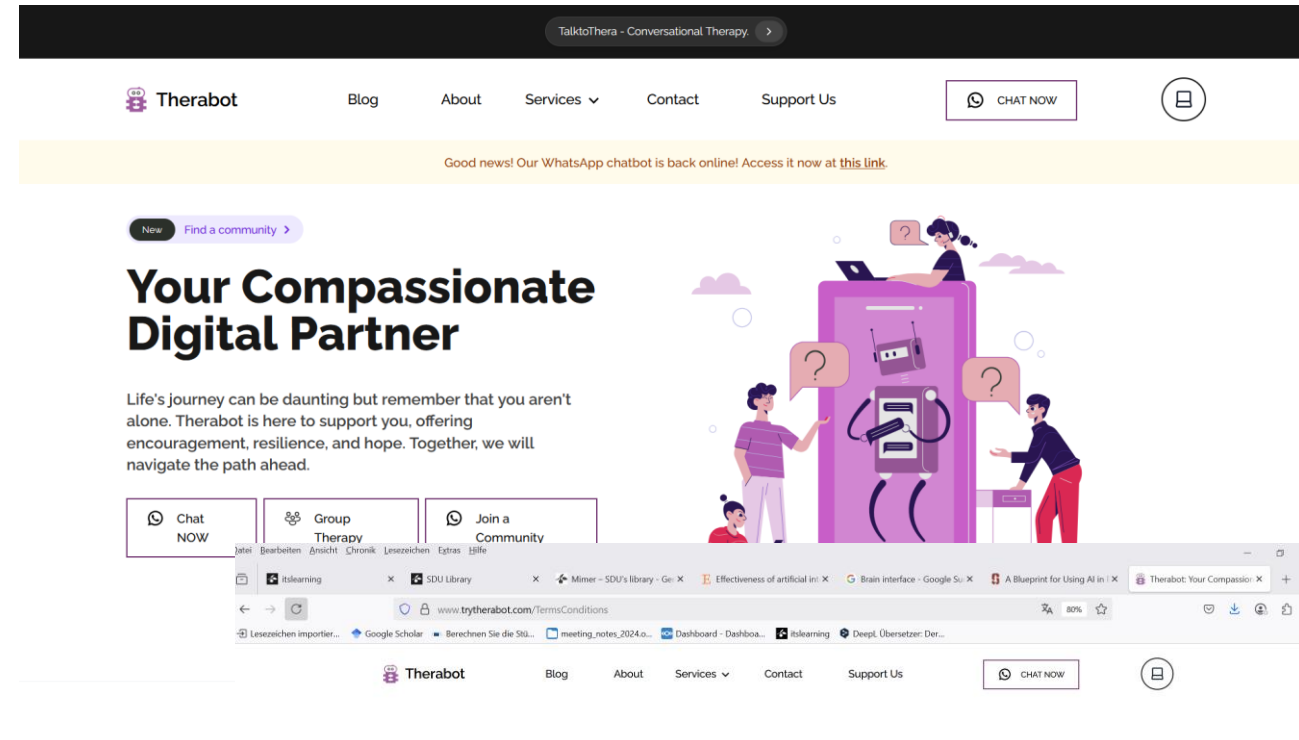
(Beg et al., 2024; Li et al., 2023)

- Several attempts to use computer-assisted digital therapy show promising results in a reduction of depression and anxiety
- However: Student samples are overrepresented, and there is still a need for more high-quality samples
- Many studies lack active control groups, focus on rather limited time frames
- big pre-registered cohort experiments are missing

Enhancement
not
Replacement

Can it be made better?

- Therabot trained by researchers at Dartmouth colleague (took them five years btw.)
- Tested with only (!) 210 patients (half of them with major depressions, general anxiety or high risk for eating disorders)
- The test lasted only 4 weeks, but it could show reductions of self-reported symptoms
- Reason enough to make it publicly available?



<https://www.trytherabot.com/TermsConditions>



The page you were looking for doesn't exist. head to the [Homepage](#)

AI in medicine

Hopes are high

Succes med kunstig intelligens på hospital i Køge skal breddes ud til hele landet, siger Løhde til Berlingske.

Sophie Løhde: Kunstig intelligens skal løse personalemangel på hospitaler



Kunstig intelligens skal fremover være med til at afhjælpe manglen på sundhedspersonale, mener sundhedsminister Sophie Løhde (V). Foto: Anders Holst Pedersen

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DANMARK

14. AUGUST 2023, 23.41

<https://politiken.dk/danmark/art9476964/Kunstig-intelligens-skal-l%C3%B8se-personalemangel-p%C3%A5-hospitaler>



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tomorrow belongs to those who embrace it today

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What if AI ran ER triage? Here's how it sped up patient care in real-world tests

A first-of-its-kind study from Yale and Johns Hopkins tested an AI program at three emergency rooms to see if it could help nurses make quicker, more accurate decisions and speed up patient flow. Here's what happened.



Written by **Tiernan Ray**, Senior Contributing Writer
March 14, 2025 at 9:46 a.m. PT
Reviewed by Elyse Betters Picaro



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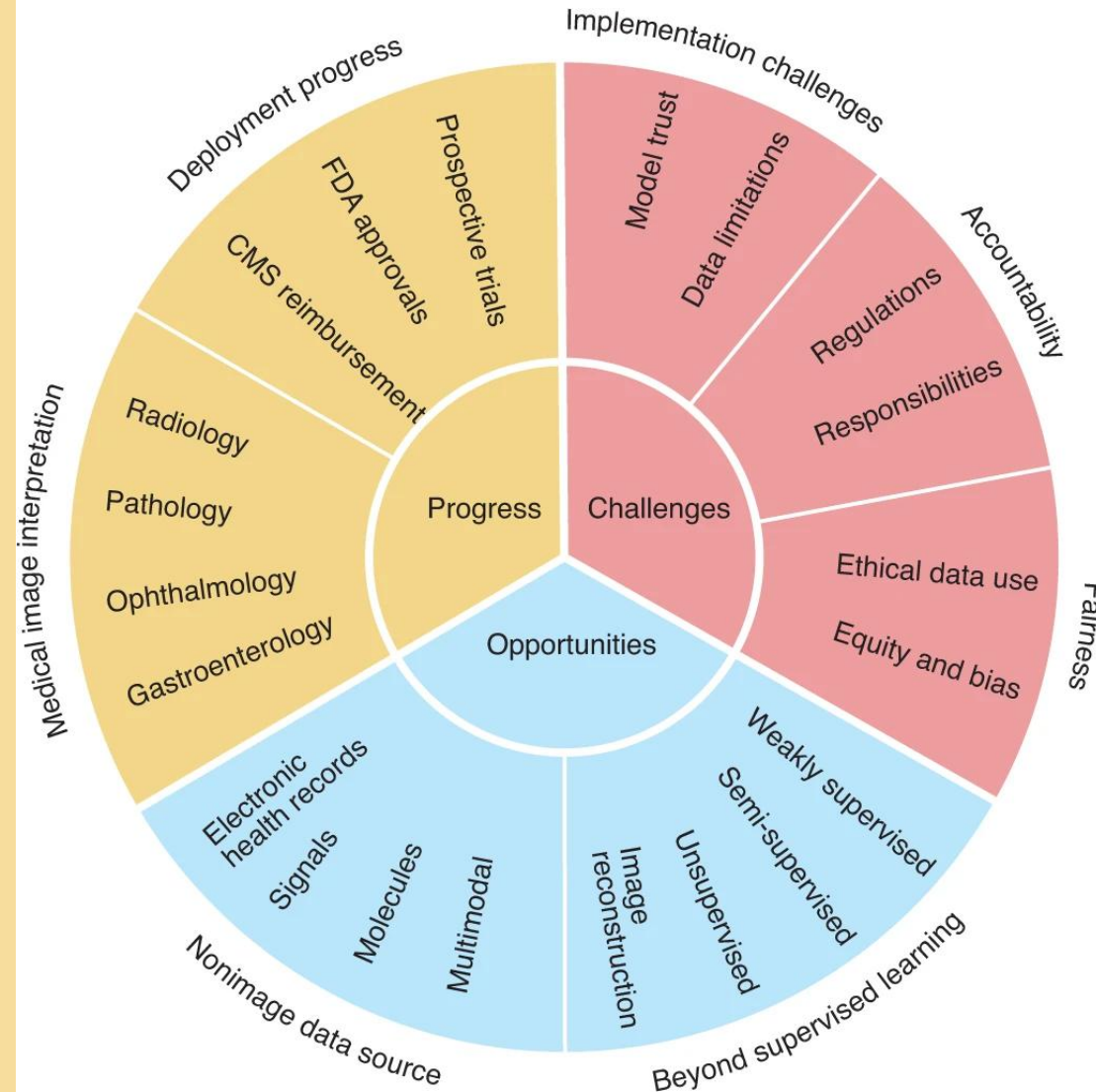
<https://www.zdnet.com/article/what-if-ai-ran-er-triage-heres-how-it-spaced-up-patient-care-in-real-world-tests/>

AI in medicine: an overview

(Rajpurkar et al., 2022; Yoon et al., 2023)

Big steps forwards in medical image analytics

- Radiology
- Pathology
- Ophthalmology
- Gastroenterology
- E.g., mammography: AI is even slightly better without humans ($\Delta_{AUC} = .05 \text{ } -.11$)
- Remember: All perception is construction (→#2 Perception)



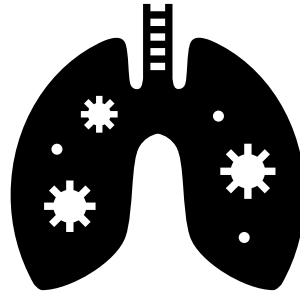
Big steps forwards in the deployment process

- Prospective randomized control trials
- Approvals by official institutions (e.g., FDA)
- Reimbursement by insurance

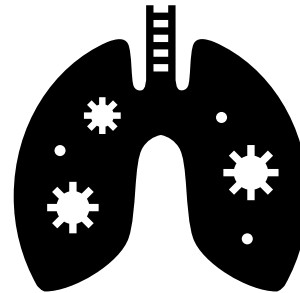
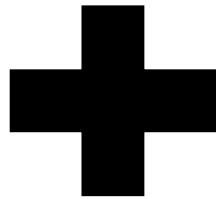
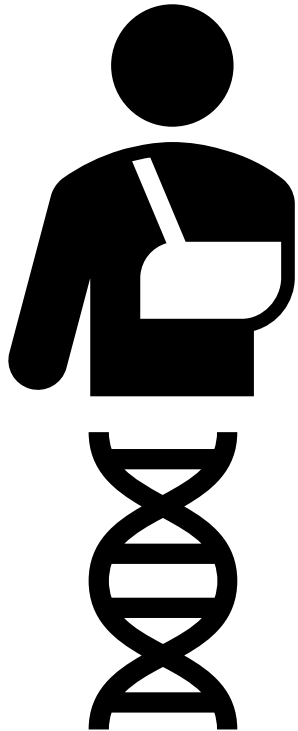
For example: the promise of personalisation

(Nationalt Genom Centre, *n.d.*)

From



To



AI can support personalised medicine

(Ghebrehiwet et al., 2024)

- BUT: highly sensitive and personalized data
- One approach: simulated data – a study based on 7000 real data showed that cGANs can be used to simulate patient data with specific characteristics and can recapitulate clinical scoring and results from clinical trials
- Basic idea: Imputing missing values in a data set using cGANs and then use the imputed data
- Another approach: Anonymize patient data using GANs – here the idea is to mimick the original data set closely while ensuring that no single combination of values can identify a specific person
- AI can furthermore improve prediction for individual patients – for example, through predicting better how they will respond to individual drugs

AI in medicine: an overview

(Ghebrehiwet et al., 2024; Rajpurkar et al., 2022; Yoon

et al., 2023)

Big challenges

- Restricted generalisability
- Lack of external evaluation

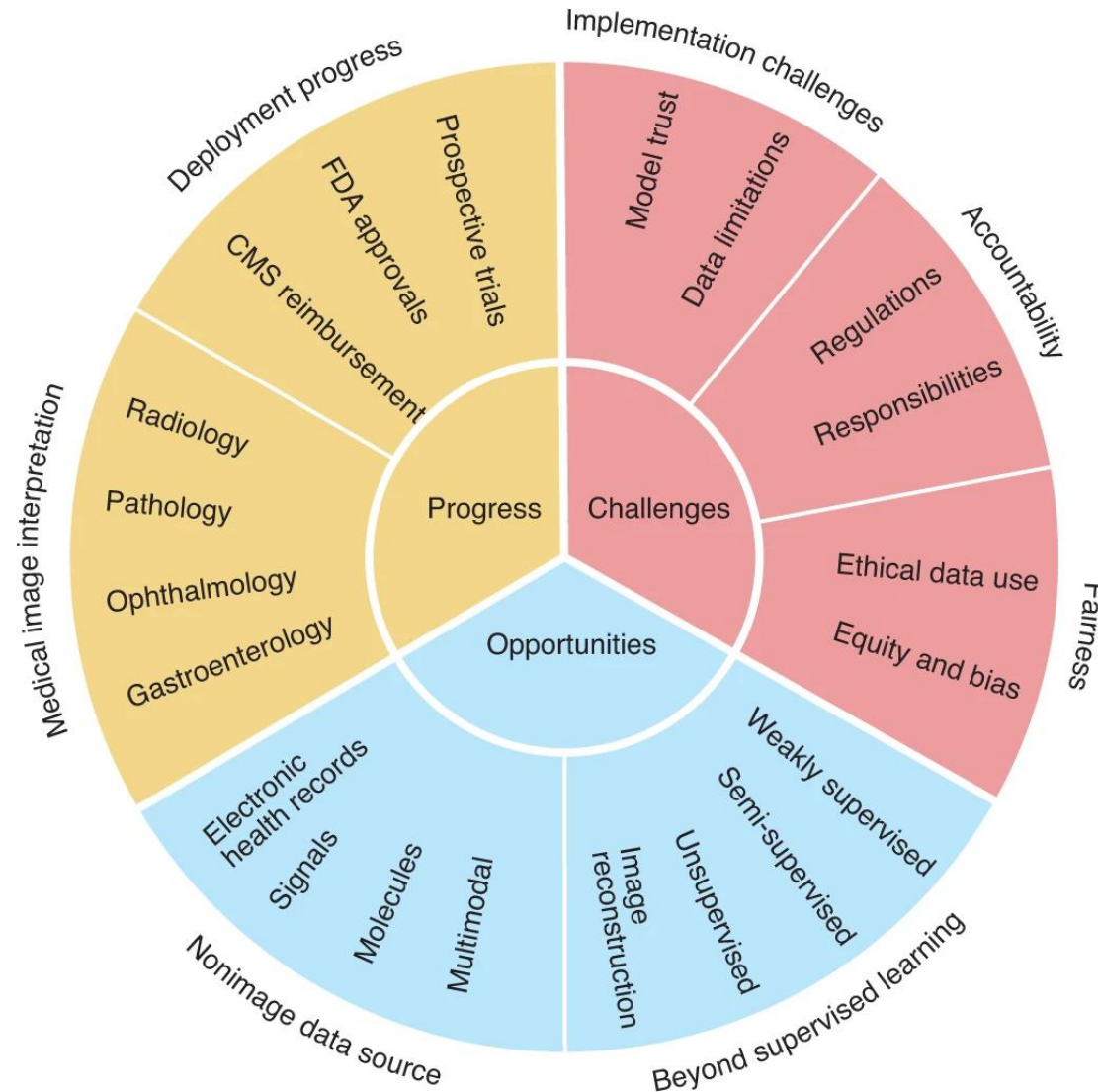
Quality & Fairness

- Bias (e.g., interpreting a ruler as a cancer cue)
- Ethical use

Accountability

- Regulations
- Responsibilities

Whom are you going to sue, if your beloved one dies?



Big challenges

- Trust in the models
 - Hallucination
 - Lack of accuracy
 - False positives
- Limitations of available data (e.g., often trained on data by one provider and on highly processed samples of tissue) → not necessary how it looks in real life

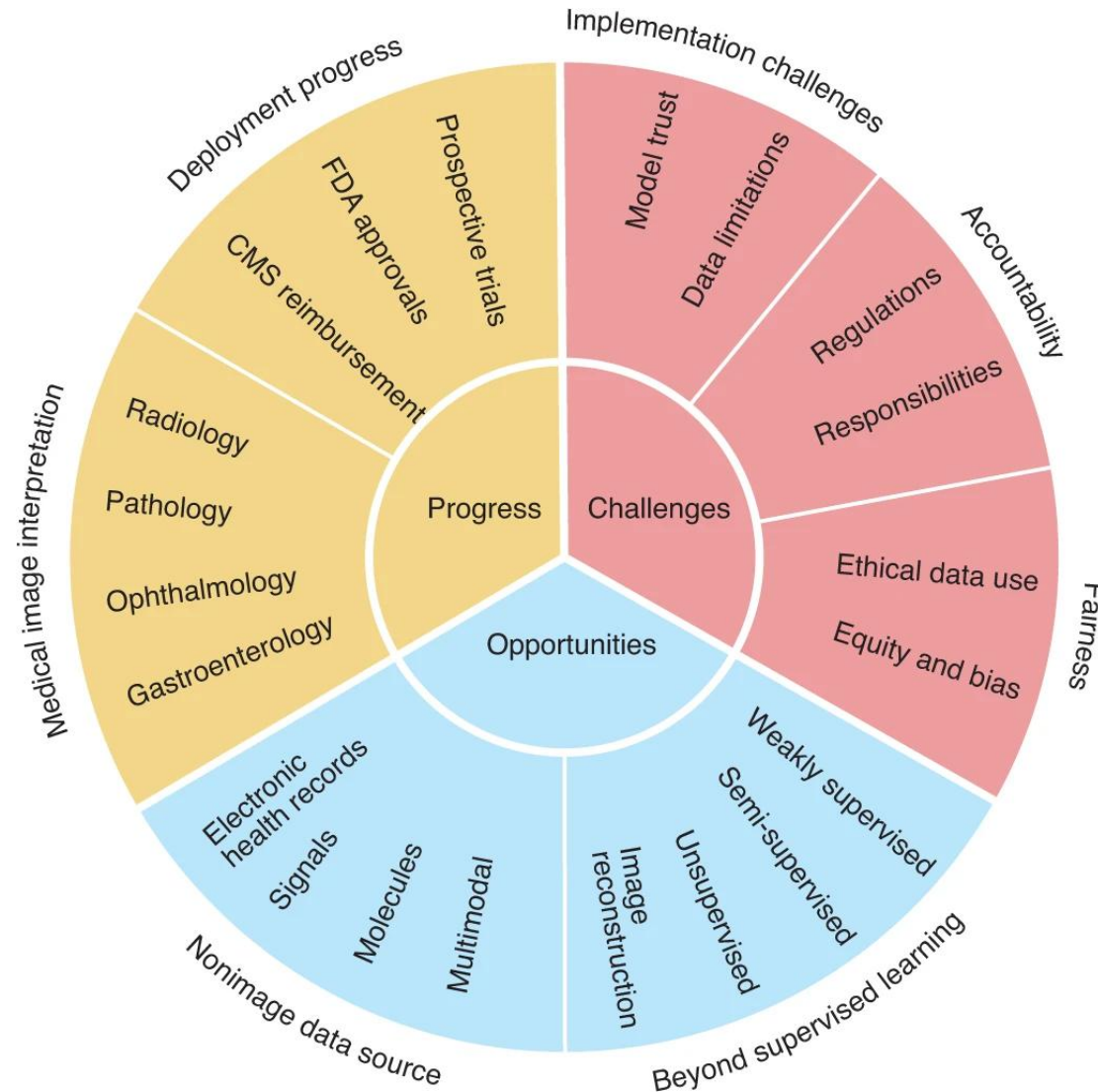
AI in medicine: an overview

(Rajpurkar et al., 2022; Yoon et al., 2023)

Big opportunities

Beyond supervised learning

- Weakly supervised learning
- Semi-supervised human in the loop approaches
- Unsupervised
- Deep-learning



Non-image data

- Electronic health records
- Natural language processing
- Signals (e.g., EEG)
- Molecules
- Multimodal data
- Genomics → e.g., informing CRISPR editing

What are your questions?



Then I hope by now

You have...

- ☐ ... a better understanding of psychological well-being
- ☐ ... insights into the relationship between AI and wellbeing
- ☐ ...discussed about AI's risks for mental health
- ☐ ... learned how AI can (not) revolutionize psychotherapy and medicine

AI 507: Artificial Intelligence and Society



#Next week: Entertainment & games with Modl.AI

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